

Software Testing — Complete Exam Ready Notes

1. Functional Testing

Definition

Functional Testing is a type of software testing that verifies whether each function or feature of a software system works according to the specified customer requirements.

বাংলা Meaning

Software-এর প্রতিটি function বা feature customer-এর requirement অনুযায়ী সঠিকভাবে কাজ করছে কি না তা পরীক্ষা করাই Functional Testing।

Key Points

- It focuses on **what the software does**.
- It verifies individual functions and features.
- It is mainly based on **customer requirements**.
- It can be performed manually or using automation.
- It is generally performed before non-functional testing.

Example

In a banking application, checking whether the **Transfer Money** function correctly transfers money from Account A to Account B.

Remember: Functional = WHAT the system does

2. Non-Functional Testing

Definition

Non-Functional Testing verifies how well a software system performs under different conditions, including performance, usability, reliability, and security.

বাংলা Meaning

Software কী কাজ করে তার চেয়ে software কতটা ভালোভাবে কাজ করে—যেমন performance, usability, reliability ইত্যাদি পরীক্ষা করাই Non-Functional Testing।

Key Points

- It focuses on **how the system works**.
- It verifies quality attributes such as:
 - Performance

- Usability
 - Reliability
 - Security
- It is mainly based on customer expectations and quality requirements.
- It is generally more difficult to perform manually.

Example

Checking whether a banking application can handle **10,000 users simultaneously without crashing**.

Remember: Non-Functional = HOW WELL it works

3. Alpha Testing

Definition

Alpha Testing is performed at the developer's site near the end of the development phase to identify defects before the software is released for Beta Testing.

বাংলা Meaning

Software development শেষ হওয়ার পর এবং Beta Testing-এর আগে developer-এর নিজস্ব environment-এ যে testing করা হয় তাকে Alpha Testing বলে।

Key Points

- Performed at the **developer's site**.
- Performed before Beta Testing.
- Main objective is to find as many defects as possible.
- It helps improve the software before external release.

Example

A software company tests a new banking application internally before giving it to customers.

Remember: Alpha = Developer's Site

4. Beta Testing

Definition

Beta Testing is a formal testing process performed by selected end-users in a real environment before the software is released to the wider market.

বাংলা Meaning

Software final release-এর আগে selected real users-এর মাধ্যমে real environment-এ testing করাকে Beta Testing বলে।

Key Points

- Performed in a **real environment**.
- Performed by selected end-users/customers.
- User feedback is collected.
- Major problems are fixed before the final release.

Example

A company gives its new mobile banking app to 500 selected customers and collects their feedback before worldwide release.

Remember: Beta = Real Users

5. Acceptance Testing / UAT

Definition

Acceptance Testing or User Acceptance Testing (UAT) verifies whether the complete system satisfies the client's business requirements and end-user needs before production release.

বাংলা Meaning

Client-এর business requirements এবং end-user needs অনুযায়ী পুরো system ঠিকভাবে কাজ করছে কি না এবং production-এ নেওয়ার জন্য client systemটি accept করবে কি না তা পরীক্ষা করা।

Key Points

- It is performed near the **final stage of testing**.
- Usually performed by the client or customer representatives.
- It checks the complete **end-to-end business flow**.
- Successful completion leads to acceptance of the system.

Example

A bank client checks account opening, fund transfer, loan processing and reporting before accepting the banking system.

Remember: UAT = Client's Final Acceptance

6. Ad-hoc Testing

Definition

Ad-hoc Testing is an informal testing approach performed without predefined test cases, formal planning, or documentation.

বাংলা Meaning

কোনো নির্দিষ্ট plan বা test case অনুসরণ না করে informalভাবে software-এর বিভিন্ন অংশ পরীক্ষা করে defect খুঁজে বের করাকে Ad-hoc Testing বলে।

Key Points

- No formal test cases.
- No predefined plan.
- Can be performed by anyone in the project.
- Tester tries different flows to find unexpected defects.

Example

A tester randomly clicks buttons, opens different menus, and enters unusual values to see whether the application crashes.

Remember: Ad-hoc = No Formal Plan

7. Accessibility Testing

Definition

Accessibility Testing verifies whether software can be effectively used by people with disabilities, including visually, hearing, physically, or cognitively impaired users.

বাংলা Meaning

Disabled বা special-needs users software সহজে এবং কার্যকরভাবে ব্যবহার করতে পারে কি না তা পরীক্ষা করা।

Key Points

It may check:

- Font size
- Color contrast
- Screen reader support
- Keyboard navigation

- Accessibility of controls

Example

Checking whether a website can be properly used by a blind user through a screen reader.

Remember: Accessibility = Can everyone use it?

8. GUI Testing

Definition

GUI Testing verifies whether the graphical user interface conforms to business requirements, detailed design documents, and approved mock-up screens.

বাংলা Meaning

Software-এর graphical interface design এবং requirements অনুযায়ী ঠিক আছে কি না তা পরীক্ষা করা।

Key Points

It checks:

- Button size and position
- Input fields
- Text alignment
- Menus
- Fonts
- Screen layout
- Navigation behavior

Example

Checking whether the Login button, textbox, menu and text alignment match the approved design.

Remember: GUI = Look + Interaction

9. Compatibility Testing

Definition

Compatibility Testing verifies whether software works correctly in different hardware, operating systems, browsers, networks, servers, and database environments.

বাংলা Meaning

Different hardware, OS, browser, network, server এবং database environment-এ software সঠিকভাবে কাজ করে কি না তা পরীক্ষা করা।

Key Points

- Tests different environments and configurations.
- Important for web and distributed applications.
- Includes browser and backward compatibility.

Example

Testing a web application on Windows, Android, Chrome, Firefox and Edge.

Remember: Compatibility = Different Environment

9.1 Browser Compatibility Testing

Definition

Browser Compatibility Testing checks whether a web application works correctly across different browsers and their versions.

বাংলা Meaning

Different browser এবং browser version-এ website ঠিকভাবে কাজ করে কি না তা পরীক্ষা করা।

Example

Testing a website in Chrome, Firefox, Edge and Safari.

9.2 Backward Compatibility Testing

Definition

Backward Compatibility Testing verifies whether a new version of software works correctly with older environments, file formats, or existing data.

বাংলা Meaning

নতুন software version পুরোনো environment, file format বা data-এর সাথে কাজ করে কি না তা পরীক্ষা করা।

Example

Checking whether a new version of an application can open files created by its older version.

Remember: New Software + Old Data

10. Back-end Testing

Definition

Back-end Testing verifies the database, tables, schemas, stored procedures, and data processing without directly using the graphical user interface.

বাংলা Meaning

GUI বাদ দিয়ে database-এর table, schema, stored procedure এবং stored data সরাসরি পরীক্ষা করা।

Key Points

It helps identify:

- Data loss
- Data corruption
- Incorrect data
- Database errors
- Deadlocks
- Stored procedure problems

Example

After updating a customer's address through the application, using an SQL query to verify that the correct address is stored in the database.

Remember: Back-end = Database Behind the GUI

11. Load Testing

Definition

Load Testing is a type of Non-Functional Testing that determines how much workload a system can handle while maintaining acceptable performance.

বাংলা Meaning

System কতটা expected workload নিতে পারে এবং সেই workload-এর মধ্যে performance acceptable থাকে কি না তা পরীক্ষা করা।

Key Points

- It is a **Non-Functional Testing** technique.
- Measures system performance under expected/increasing load.
- Checks response time and system capacity.
- Common tools include JMeter, LoadRunner, WebLoad, and Silk Performer.

Example

Testing whether a website can handle 10,000 concurrent users without significant performance degradation.

Remember: Load = Expected Workload

12. Install/Uninstall Testing

Definition

Install/Uninstall Testing verifies whether software can be correctly installed, upgraded, and uninstalled in different hardware, software, and operating system environments.

বাংলা Meaning

Different environment-এ software সঠিকভাবে install, upgrade এবং uninstall করা যায় কি না তা পরীক্ষা করা।

Example

Checking whether an application can be successfully installed, upgraded to a new version, and completely uninstalled from Windows.

Remember: Install → Upgrade → Uninstall

13. Equivalence Partitioning

Definition

Equivalence Partitioning is a Black Box Testing technique that divides input data into groups or partitions that are expected to produce similar results.

বাংলা Meaning

একই ধরনের result দেবে এমন input-গুলোকে কয়েকটি group-এ ভাগ করে প্রতিটি group থেকে representative value test করা।

Key Points

- It is a **Black Box Testing technique**.
- Reduces the number of test cases.
- Avoids testing every possible input.
- One or a few representative values are selected from each partition.

Example

Suppose an application accepts values from **-10 to +10**.

Partitions can be:

- Negative → -5
- Zero → 0
- Positive → 5

সব number test না করে প্রতিটি group থেকে representative value test করা যায়।

Remember: Equivalence = Divide into Groups

14. Boundary Value Testing

Definition

Boundary Value Testing checks the behavior of the software at and around the upper and lower limits of an input range.

বাংলা Meaning

Input range-এর lower এবং upper boundary-এর ঠিক আগে, boundary এবং ঠিক পরে value দিয়ে test করা।

Example

If the valid range is **1 to 500**, test:

0, 1, 2, 499, 500, 501

Why?

কারণ software-এর error অনেক সময় boundary-এর কাছাকাছি বেশি দেখা যায়।

Remember: Boundary = Test the Edges

15. Happy Path Testing

Definition

Happy Path Testing verifies the successful flow of a system using only valid and expected inputs.

বাংলা Meaning

শুধু valid input দিয়ে software-এর normal এবং successful flow পরীক্ষা করা।

Key Points

- Uses valid inputs.
- Focuses on successful execution.
- Does not focus on error or invalid conditions.

Example

Entering a correct username and password and checking whether the user successfully logs in.

Remember: Happy = Everything Goes Right

16. Gorilla Testing

Definition

Gorilla Testing involves repeatedly and intensively testing one specific module or functionality to verify its robustness.

বাংলা Meaning

একটি নির্দিষ্ট module বা functionality-কে খুব বেশি এবং বারবার test করে তার strength বা robustness যাচাই করা।

Example

Testing the Login module thousands of times using different valid and invalid inputs.

Remember: Gorilla = One Module, Heavy Testing

17. Monkey Testing

Definition

Monkey Testing is performed by randomly providing inputs and performing random actions without following scripted test cases.

বাংলা Meaning

কোনো predefined test case না মেনে random input ও random action দিয়ে application test করা।

Main Objective

Random data বা action-এর কারণে application crash করে কি না তা খুঁজে বের করা।

Example

Randomly typing numbers, letters and symbols into different fields and clicking random buttons.

Remember: Monkey = Random Testing

18. Incremental Integration Testing

Definition

Incremental Integration Testing integrates and tests software modules gradually rather than integrating all modules at once.

বাংলা Meaning

সব module একসাথে integrate না করে একে একে বা ছোট group করে integrate এবং test করা।

Key Points

- Modules are added gradually.
- Each integration is tested.
- Helps identify integration defects early.

Example

First integrate **Login + Registration**, test them, then add **Payment**, and test again.

Remember: Incremental = Add → Integrate → Test

19. Mutation Testing

Definition

Mutation Testing is a White Box Testing technique in which small changes are intentionally introduced into the source code to check whether existing test cases can detect the resulting defects.

বাংলা Meaning

Source code-এ ইচ্ছা করে ছোট পরিবর্তন করে দেখা হয় existing test cases সেই defect ধরতে পারে কি না।

Key Points

- It is a **White Box Testing technique**.
- Small code changes are called mutations.
- Tests the effectiveness of existing test cases.

Example

Changing:

```
if (age > 18)
```

to:

```
if (age >= 18)
```

Then checking whether existing test cases detect the change.

Remember: Mutation = Intentionally Create a Small Defect

20. Negative Testing

Definition

Negative Testing verifies whether the software correctly handles invalid, incorrect, unexpected, or unacceptable inputs.

বাংলা Meaning

ইচ্ছা করে ভুল বা invalid input দিয়ে দেখা হয় system সঠিকভাবে error handle করে কি না।

Key Points

- Tester follows an **“attitude to break”**.
- Invalid inputs are intentionally provided.
- Checks error messages and unexpected behavior.

Example

Entering -25 or ABC in an Age field and checking whether the system rejects the input with an appropriate error message.

Remember: Negative = Try to Break with Wrong Input

21. Security Testing

Definition

Security Testing verifies whether the software is protected against unauthorized access, attacks, malicious programs, and internal or external security threats.

বাংলা Meaning

Software-এর data ও system unauthorized access, hacking এবং বিভিন্ন security threat থেকে নিরাপদ কি না তা পরীক্ষা করা।

Key Points

It checks:

- Authentication
- Authorization
- Data security
- Unauthorized access
- Security vulnerabilities

Example

Checking whether a normal user can access another user's private account information.

Remember: Security = Protect Data and Access

22. Stress Testing

Definition

Stress Testing evaluates how a system behaves when it is subjected to workloads beyond its specified capacity and determines when and how the system fails.

বাংলা Meaning

System-এর specification বা capacity-এর বাইরে অতিরিক্ত চাপ দিয়ে system কখন এবং কীভাবে fail করে তা পরীক্ষা করা।

Example

A system designed for 10,000 users is tested with 20,000 users to identify its failure point.

Remember: Stress = Beyond Capacity

23. Volume Testing

Definition

Volume Testing evaluates system performance and response time when the system processes a very large volume of data.

বাংলা Meaning

অনেক বিশাল পরিমাণ data থাকলে system-এর performance ও response time কেমন থাকে তা পরীক্ষা করা।

Example

Testing a database with **10 million customer records**.

Remember: Volume = Huge Data

24. Recovery Testing

Definition

Recovery Testing verifies whether a software system can recover properly after a crash, failure, network interruption, or other disaster.

বাংলা Meaning

System crash বা কোনো failure-এর পর software কতটা ভালোভাবে আবার normal operation-এ ফিরে আসতে পারে তা পরীক্ষা করা।

Example

Disconnecting the network during a file download and checking whether the download resumes after reconnecting.

Remember: Recovery = Failure → Recover

★ 25. Smoke Testing

Definition

Smoke Testing is a preliminary testing process performed on a new software build to verify that the major and critical functions work and that the build is stable enough for further testing.

বাংলা Meaning

নতুন build পাওয়ার পর প্রথমে critical functionality-গুলো ঠিক আছে কি না পরীক্ষা করা হয়, যাতে বোঝা যায় buildটি further testing-এর জন্য acceptable কি না।

Key Points

- Performed on a **new build**.
- Initial/basic testing.
- Checks major functionality.
- Finds **show-stopper/critical defects**.
- If major problems are found, the build may be rejected.

Example

After receiving a new banking build, testers first check Login, Logout, and Money Transfer.

Remember:

Smoke = “Is the new build alive and testable?”

★ 26. Sanity Testing

Definition

Sanity Testing is a focused testing process performed on a stable build after minor changes or bug fixes to verify that the specific functionality works correctly.

বাংলা Meaning

Stable build-এ কোনো specific functionality-তে change বা bug fix করার পর মূলত ওই functionality ঠিক হয়েছে কি না পরীক্ষা করা।

Key Points

- Usually performed after a minor change or bug fix.
- Focuses on a specific area.
- Narrower than regression testing.

- Ensures the recent change works properly.

Example

A bug in the Money Transfer feature is fixed. Testers test the Transfer functionality to verify the fix.

Remember:

Sanity = “Is the specific fix/change okay?”

★ 27. Regression Testing

Definition

Regression Testing verifies that changes or bug fixes have not introduced new defects or broken previously working functionalities.

বাংলা Meaning

Software-এর কোনো অংশে change করার পর সেই change-এর কারণে আগে ঠিকভাবে কাজ করা অন্য functionality নষ্ট হয়েছে কি না তা পরীক্ষা করা।

Key Points

- Performed after modifications or bug fixes.
- Tests previously working functionality.
- Often involves a large number of test cases.
- Frequently automated because it can be repetitive and time-consuming.

Example

After modifying the Money Transfer module, testers again test Login, Balance, Statement, Payment, and Transfer.

Remember:

Regression = “Did the change break anything old?”

★ Smoke vs Sanity vs Regression

এটা exam-এর জন্য খুব important:

Feature	Smoke Testing	Sanity Testing	Regression Testing
Main purpose	Check new build	Check specific fix/change	Check old functionality

Feature	Smoke Testing	Sanity Testing	Regression Testing
Build	New build	Stable build	Modified system
Scope	Broad but shallow	Narrow	Broad
Example	Check major functions	Check fixed module	Re-test existing system

Easy Memory:

Smoke → New Build

Sanity → Specific Fix

Regression → Existing Functions

28. Static Analysis

Definition

Static Analysis evaluates software artifacts such as source code, design documents, and requirements without executing the software.

বাংলা Meaning

Software execute না করেই source code, design document এবং requirements examine করে defect খুঁজে বের করা।

Example

A developer reviews source code to find coding errors without running the program.

Remember: Static = No Execution

29. Dynamic Analysis

Definition

Dynamic Analysis evaluates the actual behavior, performance, and output of software by executing the program.

বাংলা Meaning

Software run করে তার actual behavior, output এবং performance পরীক্ষা করা।

Example

Running a banking application and checking its response time and runtime errors.

Remember: Dynamic = Execute and Check

★ Static vs Dynamic Analysis

Static Analysis

Software is not executed

Examines code/documents

Example: Code review

Dynamic Analysis

Software is executed

Examines actual behavior

Example: Running the application

Easy Trick:

Static = Read

Dynamic = Run

30. Test Driven Development (TDD)

Definition

Test Driven Development (TDD) is a software development approach in which automated tests are written before the production code.

বাংলা Meaning

Actual production code লেখার আগে automated test লিখে development করার approach হলো TDD।

Main Steps

1. Write Test → 2. Fail → 3. Write Code → 4. Pass → 5. Refactor

এটাকে বলা হয়:

Red → Green → Refactor

- **Red:** Test লিখে run করলে fail করবে।
- **Green:** Test pass করার জন্য minimum code লেখা হয়।
- **Refactor:** Code-এর quality improve করা হয়, কিন্তু functionality পরিবর্তন করা হয় না।

Example

First write a test for a `calculateTotal()` function. The test fails because the function does not exist. Then write the code, make the test pass, and finally refactor the code.

Remember: Test First → Code Later

31. Test Harness

Definition

A Test Harness is a collection of software tools and supporting components used to execute automated tests and report their results.

বাংলা Meaning

Automated test run করা এবং test-এর result report করার জন্য ব্যবহৃত tools ও supporting components-এর ব্যবস্থা হলো Test Harness।

Example

A test framework automatically runs 500 automated test cases and reports which tests passed or failed.

Remember: Test Harness = Run + Report Tests

32. Continuous Integration (CI)

Definition

Continuous Integration is a development practice in which developers frequently integrate their code into a shared repository, after which the system is automatically built and tested.

বাংলা Meaning

Developer repository-তে code update করার পর system automatically build এবং test করে দ্রুত defect detect করার practice হলো CI।

Key Points

- Developers frequently integrate code.
- CI server automatically rebuilds the system.
- Automated tests are executed.
- Defects are detected early.
- Other developers can be informed about problems quickly.

Example

Developer commits code → CI Server → Build → Automated Tests → Result

Remember: CI = Integrate → Build → Test

33. Incremental TDD for Legacy Code

Definition

Incremental TDD is an approach where automated tests are gradually added to legacy software whenever existing code is modified.

বাংলা Meaning

পুরোনো বা legacy system-এর সব test একবারে তৈরি না করে, code change করার সময় ধাপে ধাপে test যোগ করা।

Why is it needed?

Legacy systems often have:

- Few or no automated tests
- Old requirements
- Poor documentation
- Missing or outdated design

তাই পুরো system-এর জন্য একসাথে test তৈরি করা expensive এবং difficult।

Example

A developer needs to modify the old Payment module. Before modifying it, the developer adds automated tests for the affected functionality and then changes the code.

Remember:

Legacy Code → Modify → Add Test → Refactor

34. User Story

এটি Testing Type নয়, কিন্তু Agile Testing-এর গুরুত্বপূর্ণ concept।

Definition

A User Story is a short description of a software requirement written from the perspective of the end-user.

বাংলা Meaning

End-user software দিয়ে কী করতে চায় তা user-এর perspective থেকে ছোট করে প্রকাশ করাকে User Story বলে।

Example

“As a customer, I want to transfer money so that I can send money to another account.”

Remember: User Story = What the User Wants

Question Answer:

c) Software Testing Types & Differences

- i) Functional and non-functional testing.
- ii) Alpha testing and Beta testing.
- iii) Load testing and stress testing.
- iv) Differentiate between Black box testing and white box testing.
- v) Differentiate between smoke testing, sanity testing, and regression testing.
- vi) Differentiate between monkey and gorilla testing.

c) Software Testing Types & Differences

i) Functional Testing vs. Non-Functional Testing

Functional Testing

Definition:

Functional Testing is a type of software testing that verifies whether the software performs its specified functions correctly according to the customer's requirements.

Example:

Checking whether a user can successfully log in using a valid username and password.

Non-Functional Testing

Definition:

Non-Functional Testing verifies how well the software performs under different conditions, such as performance, usability, reliability, and security.

Example:

Checking whether the application can support 10,000 users simultaneously without crashing.

Differences

Basis	Functional Testing	Non-Functional Testing
1. Focus	Checks what the system does.	Checks how well the system works.
2. Purpose	Verifies functions and features.	Verifies quality attributes such as performance, usability, reliability, and security.
3. Basis	Mainly based on specified customer requirements.	Mainly based on quality requirements and customer expectations.
4. Main Concern	Correctness of functionality.	Quality and performance of the system.
5. Example	Checking whether money transfer works correctly.	Checking whether money transfer works within 2 seconds under heavy load.
6. Execution	Can often be performed manually or automatically.	Often requires specialized tools and is more difficult to perform manually.

Easy Memory

Functional = WHAT it does

Non-Functional = HOW WELL it works

ii) Alpha Testing vs. Beta Testing**Alpha Testing****Definition:**

Alpha Testing is performed at the developer's site near the end of the development phase to identify defects before the software is released to external users.

Example:

A software company's testing team tests a new banking application internally before giving it to customers.

Beta Testing

Definition:

Beta Testing is performed by selected end-users in a real environment before the final release of the software.

Example:

A company provides its new banking application to selected customers to use in real life and provide feedback.

Differences

Basis	Alpha Testing	Beta Testing
1. Location	Usually performed at the developer's site .	Performed in a real user environment .
2. Performed By	Developers, testers, or internal staff.	Selected end-users or customers.
3. Stage	Performed before Beta Testing.	Performed after Alpha Testing and before final release.
4. Environment	Controlled/internal environment.	Real-world environment.
5. Main Objective	Find and fix as many defects as possible.	Identify real-world problems and collect user feedback.
6. Users	Internal users.	External/selected real users.
7. Feedback	Mainly technical/internal feedback.	Mainly real-user experience and feedback.

Easy Memory

Alpha → At developer's site

Beta → By real users

★ One-Line Difference

Alpha testing is internal testing at the developer's site, whereas Beta testing is testing by selected end-users in a real environment.

iii) Load Testing vs. Stress Testing

Load Testing

Definition:

Load Testing is a type of Non-Functional Testing that evaluates the performance and capacity of a system under the expected or specified workload.

Example:

Testing a website with 10,000 concurrent users when 10,000 users is the expected workload.

Stress Testing**Definition:**

Stress Testing evaluates how a system behaves when it is subjected to workloads beyond its specified capacity and determines when and how the system fails.

Example:

Testing the same website with 20,000 users when its specified capacity is 10,000 users.

Differences

Basis	Load Testing	Stress Testing
1. Purpose	Checks performance under expected workload.	Checks system behavior beyond its capacity.
2. Workload	Uses normal or expected workload.	Uses extreme or abnormal workload.
3. Main Concern	Performance and capacity.	Failure point and system stability.
4. Objective	Determines whether the system can handle the expected load efficiently.	Determines when and how the system fails under excessive load.
5. Example	Testing with 10,000 users.	Testing with 20,000 users when capacity is 10,000.
6. Result	Measures response time, throughput, and performance.	Observes failure, degradation, recovery, and stability.

Easy Memory

Load = How much can it handle?

Stress = How much before it breaks?

★ One-Line Difference

Load Testing checks system performance under expected workload, whereas Stress Testing pushes the system beyond its capacity to find its failure point.

iv) Black Box Testing vs. White Box Testing

Black Box Testing

Definition:

Black Box Testing is a testing technique in which the internal structure or source code of the software is not known or considered by the tester.

বাংলা অর্থ:

ভেতরের code বা internal structure না দেখে শুধু input এবং expected output-এর ভিত্তিতে software test করা।

Example:

Entering a valid username and password and checking whether login is successful without looking at the login code.

White Box Testing

Definition:

White Box Testing is a testing technique in which the internal structure, source code, logic, and implementation of the software are known and examined by the tester.

বাংলা অর্থ:

Software-এর internal code, logic এবং structure জেনে বা দেখে testing করা।

Example:

Checking whether every branch and condition in a login function has been executed and tested.

Differences

Basis	Black Box Testing	White Box Testing
1. Internal Code	Internal code is not considered.	Internal code is examined.
2. Knowledge Required	No detailed programming knowledge is required.	Requires programming and code knowledge.
3. Focus	Focuses on input, output, and functionality.	Focuses on code, logic, paths, and internal structure.

Basis	Black Box Testing	White Box Testing
4. Tester Perspective	Tests software from the user's perspective .	Tests software from the developer's/internal perspective .
5. Test Basis	Requirements and specifications.	Source code and program logic.
6. Common Techniques	Equivalence Partitioning, Boundary Value Testing, etc.	Statement, branch, path, and mutation testing.
7. Example	Test login with valid/invalid credentials.	Test every condition and branch inside the login code.

Easy Memory

Black Box = Don't see inside

White Box = See inside

★ One-Line Difference

Black Box Testing tests the external behavior without examining the internal code, whereas White Box Testing tests the internal logic and source code of the software.

v) Smoke Testing vs. Sanity Testing vs. Regression Testing

★ এই প্রশ্নটা খুব important | Table-এ দিলে examiner-এর কাছে খুব clear হবে।

Smoke Testing

Definition:

Smoke Testing is a preliminary test performed on a new build to verify that the major and critical functions work and that the build is stable enough for further testing.

Example:

After receiving a new banking application build, testers check Login, Logout, and Money Transfer first.

Remember:

Smoke = New Build Check

Sanity Testing

Definition:

Sanity Testing is a focused test performed on a stable build after a minor change or bug fix to verify that the specific functionality works correctly.

Example:

After fixing a Money Transfer bug, testers check the Transfer functionality to verify the fix.

Remember:

Sanity = Specific Fix Check

Regression Testing**Definition:**

Regression Testing verifies that recent changes or bug fixes have not affected previously working functionalities of the software.

Example:

After modifying Money Transfer, testers check Login, Balance, Statement, Payment, and other existing features again.

Remember:

Regression = Old Function Check

Differences

Basis	Smoke Testing	Sanity Testing	Regression Testing
1. Main Purpose	Checks whether a new build is stable enough for testing.	Checks whether a specific change or bug fix works correctly.	Checks whether existing features are still working after changes.
2. Performed On	Usually a new build .	A stable build after minor changes/fixes.	A modified system after changes or bug fixes.
3. Scope	Broad but shallow.	Narrow and focused.	Broad and relatively comprehensive.
4. Focus	Critical/basic functionality.	Specific modified functionality.	Previously working functionalities.
5. Objective	Detect major/show-stopper defects.	Verify the correctness of a particular fix/change.	Detect side effects caused by changes.
6. Example	Check Login, Logout, Transfer in a new build.	Check only fixed Transfer feature.	Re-test Login, Transfer, Balance, Statement, etc.

Basis	Smoke Testing	Sanity Testing	Regression Testing
7. Nature	Preliminary testing.	Focused testing.	Repeated/comprehensive testing.

🔥 Best Memory Trick

Imagine a **new banking software build** arrives:

Step 1 — SMOKE:

☞ “Is this build alive?”

Check major features.

Step 2 — SANITY:

☞ “The Transfer bug was fixed. Is that fix okay?”

Check the specific area.

Step 3 — REGRESSION:

☞ “Did fixing Transfer break Login, Balance or Statement?”

Check previously working features.

★ Perfect One-Line Answer

Smoke testing checks whether a new build is stable, Sanity testing checks whether a specific change or fix works correctly, and Regression testing checks whether the change has broken any previously working functionality.

vi) Monkey Testing vs. Gorilla Testing

Monkey Testing

Definition:

Monkey Testing is an informal testing technique in which random inputs and random actions are performed without following predefined test cases.

বাংলা অর্থ:

কোনো নির্দিষ্ট test case বা plan ছাড়াই random input এবং random action দিয়ে software test করা।

Example:

Randomly entering letters, numbers, symbols and clicking different buttons to see whether the application crashes.

Remember:

Monkey = Random

Gorilla Testing

Definition:

Gorilla Testing is an intensive testing technique in which one specific module or functionality is tested repeatedly and heavily to verify its robustness.

বাংলা অর্থ:

একটি নির্দিষ্ট module-কে বারবার এবং খুব intensiveভাবে test করে তার robustness যাচাই করা।

Example:

Testing the Login module thousands of times with different valid and invalid inputs.

Remember:

Gorilla = One module, heavily tested

Differences

Basis	Monkey Testing	Gorilla Testing
1. Approach	Random testing.	Intensive and focused testing.
2. Test Cases	Usually no predefined test cases.	May use planned/repeated test scenarios.
3. Focus	Different parts of the application randomly.	One specific module or functionality.
4. Input	Random and unexpected inputs.	Repeated and targeted inputs.
5. Main Objective	Find unexpected crashes and hidden defects.	Check the robustness and stability of a particular module.
6. Coverage	Broad/random coverage.	Narrow/deep coverage.
7. Example	Randomly clicking buttons throughout an application.	Testing the Login module repeatedly thousands of times.

🔥 Easy Memory

🐒 **Monkey** → Randomly jumps everywhere

🦍 **Gorilla** → Attacks one place repeatedly

★ One-Line Difference

Monkey Testing randomly tests different parts of the application, whereas Gorilla Testing repeatedly and intensively tests one specific module or functionality.

